

CANTEEN OPTIMIZATION USING QUEUING THEORY AND ONLINE ORDER MANAGEMENT SYSTEM

PROJECT REPORT

submitted by

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Department of Mechanical Engineering
College of Engineering Trivandrum, Kerala

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**DEPARTMENT OF MECHANICAL ENGINEERING
COLLEGE OF ENGINEERING
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CERTIFICATE

This is to certify that the report entitled '**CANTEEN OPTIMIZATION USING QUEUING THEORY AND ONLINE ORDER MANAGEMENT SYSTEM**' submitted by '**Mr Jobin S Babu(TVE22IE036), Mr Adarsh Jayan(TVE22IE003), Mr Arjun B(TVE22IE018) and Mr Sanilesh S(TVE22IE059)**' to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Industrial Engineering is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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We, the undersigned hereby declare that the project report '**CANTEEN OPTIMIZATION USING QUEUING THEORY AND ONLINE ORDER MANAGEMENT SYSTEM**' submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of Prof. Suresh Kumar V.B. This submission represents our ideas in our own words and where ideas or words of others have been included; we have adequately and accurately, cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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ABSTRACT

Canteen management in academic institutions faces persistent challenges during peak hours, including long waiting queues, inefficient service processes, food wastage due to poor demand forecasting, and lack of centralized data management. This project addresses these challenges through a two-pronged approach: analytical queue modelling and a software-based Online Order Management System.

In the first phase, queuing theory was applied to the existing canteen system at College of Engineering Trivandrum. Real-world data on customer arrival rates and service times were collected and analysed. Using the Kingman Approximation equation, the average waiting times for three models were estimated and compared: (1) the current single-server system, (2) a two-parallel-billing model, and (3) a two-dedicated-billing model. Simulations were performed to validate the analytical results. The two-parallel-billing model was found to reduce predicted waiting time by approximately 93% compared to the current system, and outperformed the two-dedicated-billing model by approximately 68%.

In the second phase, an Online Order Management System was developed using the Python Flask framework. The system features a Menu Management module for real-time item control, a Sales Analytics dashboard for data-driven decision-making, and a Demand Forecasting module that uses historical sales data to predict demand for the upcoming day. The integrated tokenisation interface digitises the ordering process, reducing physical queuing and enabling better operational planning.

The combined approach demonstrates that applying industrial engineering principles alongside modern software tools can significantly improve canteen efficiency, customer satisfaction, and resource utilisation.

Keywords: Queuing Theory, Kingman Approximation, M/M/C Model, Order Management System, Demand Forecasting, Canteen Optimization, Flask, Simulation

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Chapter 1:

ABBREVIATIONS

OMS	Order Management System
OOMS	Online Order Management System
M/M/1	Markovian Arrival, Markovian Service, Single Server
M/M/c	Markovian Arrival, Markovian Service, Multiple Servers
M/G/1	Markovian Arrival, General Service, Single Server
G/G/m	General Arrival, General Service, m Servers
FIFO	First In First Out
VUT	Variability–Utilization–Time (Kingman Approximation)
HTML	HyperText Markup Language
CSS	Cascading Style Sheets

Chapter 2:

NOTATIONS

λ (lambda)	Arrival rate
μ (mu)	Service rate
ρ (rho)	Utilization factor
W_q	Average waiting time in queue
C_a	Coefficient of variation of inter-arrival time
C_s	Coefficient of variation of service time
m	Number of servers
σ_a	Standard deviation of inter-arrival time
σ_s	Standard deviation of service time

Chapter 3

INTRODUCTION

This chapter introduces the problem of inefficiencies in institutional canteens, particularly during peak hours, where long queues, delays, and lack of data-driven decision-making affect performance. It outlines the motivation behind the project, highlights the key issues such as food wastage and manual processes, and defines the scope of applying queuing theory along with a digital order management system to improve efficiency.

3.1 Background

Canteens in academic institutions play a vital role in supporting the daily needs of students, faculty, and staff by providing affordable and timely meals. Unlike commercial food outlets that operate under relatively distributed customer inflow, institutional canteens experience highly concentrated demand during specific time intervals, particularly during short academic breaks. These peak periods create intense pressure on the service system, often resulting in long queues, overcrowding, and reduced service efficiency.

The canteen at the College of Engineering Trivandrum is no exception to this pattern. A significant surge in customer arrivals occurs within a limited time window, making queue management a critical operational concern. Since the available service capacity is often not dynamically adjusted to match the fluctuating demand, customers experience increased waiting times. This not only affects the overall dining experience but also reduces the effective time available for students to eat and rest before returning to academic activities.

Traditionally, such canteens operate using manual processes for order placement, token issuance, billing, and inventory tracking. These systems rely heavily on human intervention and lack integration, leading to inefficiencies such as delays in service, errors in order handling, and poor coordination between ordering and preparation stages. Furthermore, the absence of digital data recording prevents systematic analysis of customer demand patterns.

Without reliable data, decision-making becomes largely intuitive rather than evidence-based. As a result, canteen operators often struggle to balance food production with actual demand, leading to either overproduction (causing food wastage) or underproduction (resulting in stockouts of popular items). Therefore, improving operational efficiency through systematic analysis and digital transformation has become essential.

3.2 Problem Statement

During peak hours, the canteen experiences a high volume of customer arrivals within a short time span, leading to congestion at service counters and extended waiting times in queues. This overcrowding negatively impacts customer satisfaction and reduces the efficiency of the overall system.

The current system lacks a centralized and digitalized data management framework. All transactions are handled manually, and there is no mechanism to record or analyze sales data. Consequently, the management is unable to accurately forecast demand, identify consumption trends, or make informed decisions regarding inventory and production planning.

This absence of data-driven operations results in two major inefficiencies: significant food wastage due to overproduction and frequent stockouts of high-demand items due to underestimation. Additionally, the single-channel billing and tokening process further exacerbates delays during peak periods.

To address these challenges, this project proposes the development of an Online Order Management System integrated with queuing theory-based analysis. The system aims to reduce waiting time, improve service efficiency, and enable real-time data analytics for better decision-making.

3.3 Motivation

The motivation for undertaking this project stems from both personal observation and practical challenges faced within the institutional canteen environment. As regular users of the canteen, students frequently encounter long waiting times, especially during short breaks between classes. This often results in either skipping meals or rushing through them, negatively affecting both comfort and productivity.

Informal feedback and discussions among students further highlighted dissatisfaction with the current system, particularly regarding delays, lack of transparency in order handling, and unavailability of popular food items during

peak hours. These issues indicated the need for a systematic solution to improve both service quality and operational efficiency.

From an academic perspective, this problem presents an excellent opportunity to apply concepts from Industrial Engineering, particularly queuing theory, simulation, and operations management. The project integrates theoretical knowledge with practical implementation by analyzing real-world data and developing a technological solution.

Additionally, the increasing trend of digital transformation in service industries inspired the idea of incorporating an online order management system. By digitizing the process, it becomes possible to capture valuable data, perform analytics, and implement demand forecasting techniques. This not only enhances efficiency but also aligns the canteen operations with modern smart service systems

3.4 Scope of the Project

The scope of this project is confined to the canteen facility at the College of Engineering Trivandrum. The study focuses on analyzing the existing queuing system during peak hours and evaluating its performance using empirical data collected from real observations.

The queuing analysis is performed using theoretical models and approximations to study the impact of varying arrival rates and service mechanisms on waiting time. The project also explores alternative service configurations, such as parallel billing systems, to identify potential improvements in efficiency.

In addition to analytical modeling, a web-based Online Order Management System is developed as a prototype. The system includes features such as menu management, digital tokening, sales analytics, and demand forecasting. However, it is important to note that the system is not deployed in a live production environment and is intended primarily for demonstration and evaluation purposes.

The demand forecasting module operates using historical sales data generated within the system and applies basic predictive techniques to estimate future demand. The scope does not include advanced machine learning models or large-scale deployment considerations.

Overall, the project aims to demonstrate how the integration of queuing theory and digital systems can significantly improve operational efficiency, reduce waiting times, and enable data-driven decision-making in institutional canteens.

Chapter 4

LITERATURE REVIEW

This chapter reviews existing research related to queuing theory applications in service systems and digital food ordering solutions. It discusses various studies on single-server and multi-server models, simulation approaches, and digital interfaces, and identifies a research gap where both analytical modeling and practical system implementation are not integrated, which this project aims to address.

4.1 Overview

This chapter reviews existing literature relevant to the present study, focusing on three major domains: queuing theory in service systems, applications in canteen and restaurant environments, and digital order management solutions. These areas collectively form the theoretical and practical foundation of this project.

Queuing theory provides the mathematical framework required to analyze waiting lines, service efficiency, and system performance under varying demand conditions. Previous studies have demonstrated its effectiveness in optimizing service systems such as banks, hospitals, and food service environments. In parallel, recent advancements in digital technologies have introduced online ordering systems and self-service interfaces that significantly enhance operational efficiency and customer satisfaction.

The present project integrates both these dimensions by combining queuing analysis with the development of a digital order management system. Therefore, the reviewed literature helps in understanding both the analytical techniques and technological solutions relevant to the problem.

4.2 Queuing Theory in Service Systems

Molla (2017) conducted a case study applying classical queuing models, particularly the M/M/1 model, to analyze service performance in real-world environments such as restaurants and banking systems. The study focused on

evaluating key performance indicators such as average waiting time, queue length, and system utilization. It demonstrated that even simple queuing models can effectively identify bottlenecks and inefficiencies in service systems. This work forms the foundation for the baseline analysis of the current single-server system in this project, where similar parameters such as arrival rate, service rate, and utilization factor are evaluated.

Jiao et al. (2022) extended queuing theory applications by using the M/M/c model to determine the optimal number of service counters in a university canteen. Through simulation, the study showed how increasing the number of servers reduces waiting time but also impacts system utilization. The findings highlight the importance of balancing service capacity and demand. This research is directly relevant to the present project, where multiple service configurations, including parallel billing systems, are analyzed to reduce waiting time during peak hours.

Hidayana and Yohandoko (2024) applied the M/M/c queuing model to fast food restaurant operations, focusing on metrics such as server utilization, average queue length, and customer waiting time. Their results provided practical insights into how multi-server systems perform under high demand conditions. These findings serve as a benchmark for comparing the performance of the proposed models in this project, particularly in evaluating improvements over the existing single-server system.

Whitt (2012) provided a comprehensive theoretical framework for analyzing the M/G/1 queue, which allows for general service time distributions. The study introduced approximations, including the Kingman formula, to estimate waiting time in systems where arrival and service processes are not strictly exponential. This work is particularly important for the present project, as the Kingman approximation is used to analyze waiting times under varying arrival variability (coefficient of variation). It enables a more realistic evaluation of system performance compared to idealized models.

In addition to these, simulation-based approaches have been widely used in queuing analysis to capture real-world variability. Such methods allow for testing different system configurations without physically altering the system. This concept is applied in the present project to compare current and proposed models under different arrival scenarios

4.3 Digital Ordering and Canteen Management Systems

Kale et al. (2020) proposed an online food ordering system specifically designed for college canteens using an Android application integrated with QR code-based ordering. The system aimed to reduce physical queues and improve order processing efficiency by enabling customers to place orders digitally. The study demonstrated that digital ordering systems can significantly streamline operations and reduce waiting time. This work provides a conceptual basis for the online tokening and order management system developed in the present project, although the current project extends it further by integrating analytics and forecasting.

Shahril et al. (2021) examined customer satisfaction in quick service restaurants with the introduction of self-service kiosks. The study found that digital interfaces improve perceived service quality by reducing waiting time, minimizing human interaction delays, and providing greater transparency in order processing. These findings strongly support the implementation of a digital ordering interface in institutional canteens, as proposed in this project. It validates the idea that technological intervention can enhance both operational efficiency and user experience.

Recent trends in digital transformation emphasize the use of data-driven systems for decision-making. Unlike traditional systems, digital platforms enable continuous data collection, which can be analyzed to identify patterns in customer behavior. The present project builds upon this concept by incorporating sales

analytics and demand forecasting modules, thereby extending beyond simple order automation to intelligent system management

4.4 Summary Table of Literature Review

Title	Author(s) & Year	Key Contribution / Relevance
M/M/1 Application in Restaurants (IOSR-JBM)	Molla (2017)	Applied M/M/1 model to real service systems; forms the basis for analyzing the current single-server system.
University Canteen Simulation with M/M/c	Jiao et al. (2022)	Identified optimal number of servers; directly supports multi-server modeling in this project.
Online Food Ordering for College Canteen	Kale et al. (2020)	Introduced digital ordering via Android and QR; conceptual basis for online tokening system.
Customer Satisfaction at Self-Service Kiosks	Shahril et al. (2021)	Demonstrated improved satisfaction with digital interfaces; supports system adoption.
M/M/c in Fast Food Restaurants	Hidayana & Yohandoko (2024)	Provided performance benchmarks for multi-server systems under high demand.
M/G/1 Queue — IEOR 6711	Whitt (2012)	Theoretical foundation for Kingman approximation used in waiting time estimation.

4.5 Research Gap

The review of existing literature indicates that significant work has been done separately in the areas of queuing theory and digital ordering systems. Queuing theory studies primarily focus on mathematical modeling and optimization of service systems, while digital system studies emphasize automation and user interface improvements.

However, there is a lack of integrated approaches that combine rigorous queuing analysis with the development of a functional digital order management system for the same application. Most studies either remain theoretical or focus solely on software implementation without analytical validation.

This project addresses this gap by combining both approaches. It not only analyzes the existing system using queuing theory and simulation techniques but also develops a practical online order management system that captures real-time data and enables demand forecasting. This integration provides a comprehensive solution to both operational inefficiencies and data management limitations in institutional canteens.

Chapter 5

OBJECTIVES

This chapter outlines the primary and secondary objectives of the project, focusing on analyzing the impact of arrival variability on waiting time and developing an Online Order Management System. It also includes goals such as data collection, application of the Kingman approximation, simulation validation, and comparison of different queue models.

5.1 Primary Objectives

The primary objectives of this project focus on addressing the core challenges identified in the existing canteen system by integrating analytical modeling with digital system development.

1. **To study the effect of variation of inter-arrival time on waiting time in the queue for the current canteen system as well as two proposed alternative service models.**

This objective aims to understand how fluctuations in customer arrival patterns influence system performance, particularly during peak hours. By analyzing different arrival scenarios, the study evaluates the sensitivity of waiting time to demand variability and identifies optimal service configurations.

2. **To develop an Online Order Management System capable of recording sales data and enabling demand forecasting and real-time analytics.**

This objective focuses on digitizing the existing manual processes by creating a web-based platform that captures transaction data, improves order handling efficiency, and supports data-driven decision-making.

5.2 Secondary Objectives

The secondary objectives support the primary goals by providing the necessary analytical, technical, and implementation framework:

- To collect and analyse empirical data on customer arrival rates and service rates at the College of Engineering Trivandrum canteen.

This involves real-time observation and measurement of system parameters to ensure that the analysis reflects actual operating conditions.

- To apply the Kingman Approximation equation to estimate average waiting times under varying arrival and service conditions.

This enables the evaluation of system performance under realistic assumptions where variability exists in arrivals and service processes.

- To validate analytical results using discrete-event simulation.

Simulation is used to model real-world system behavior and verify the accuracy of theoretical results obtained from queuing models.

- To compare the performance of three queue configurations: the current system, two-parallel-billing model, and two-dedicated-billing model.

This comparison helps in identifying the most efficient configuration in terms of waiting time reduction and system utilization.

- To design and implement a web-based platform with features such as menu management, sales analytics, and demand forecasting modules.

This ensures that the system not only reduces waiting time but also improves operational efficiency through digital integration.

- To assess the feasibility of the proposed improvements in a real institutional canteen setting.

This includes evaluating whether the proposed solutions can be practically implemented considering operational constraints and user acceptance.

Chapter 6

METHODOLOGY

This chapter explains the systematic approach followed in the project, divided into two phases: queuing analysis and software development. It covers survey-based problem identification, data collection through observation, statistical processing, system modeling, simulation, and the development of a web-based order management system with various functional modules.

6.1 Overall Approach

This project adopts a structured approach combining analytical modelling and software development to improve the efficiency of the canteen system.

The methodology is divided into two major phases:

- Phase 1 — Queuing Analysis:

Involves problem identification, data collection, statistical processing, and modelling of the canteen service system.

- Phase 2 — Software Development:

Involves the design and development of an Online Order Management System to digitize operations and improve decision-making

6.2 Phase 1 — Queuing Analysis

6.2.1 Preliminary Study — Customer Survey

A preliminary survey was conducted among students of the College of Engineering Trivandrum to understand user experience and identify key issues in the existing canteen system. The survey was created using Google Forms and distributed among regular canteen users.

The questionnaire focused on factors such as waiting time, service satisfaction, peak-hour congestion, and interest in digital ordering systems. The responses indicated significant dissatisfaction due to long queues and delays, and a strong preference for a more efficient and digitalized system.

The findings from the survey helped in validating the problem and guided the direction of further analysis.

6.2.2 Data Collection

Primary data was collected through direct observation and using cctv footage of the canteen during peak hours. The objective was to capture real-time operational behaviour of the system.

The following data was recorded:

- Customer arrival times
- Inter-arrival times
- Service times

Data collection was carried out over multiple days to ensure reliability. All observations were recorded and organized using Microsoft Excel for further processing.

6.2.3 Data Processing

The collected data was processed using Microsoft Excel to obtain the necessary statistical measures required for analysis. Basic statistical parameters such as averages and variability measures were computed.

These processed values were used as inputs for the queuing models and further analysis presented in subsequent chapters.

6.2.4 System Modelling Approach

The canteen system was modelled using queuing theory concepts. The existing system and proposed alternatives were represented as different queue configurations.

The analysis focuses on comparing:

- The current single-server system
- A two-parallel-billing system

- A two-dedicated-billing system

The performance of these models is evaluated in later chapters.

6.2.5 Simulation Approach

To support the analytical study, simulation was used to replicate the behaviour of the queuing system under different conditions.

The simulation model was developed based on the collected data and was used to evaluate system performance metrics such as waiting time and queue length. The results of the simulation are discussed in a separate chapter.

6.3 Phase 2 — Software Development

6.3.1 System Development Approach

An Online Order Management System was developed to address the limitations of the existing manual system. The development followed a modular approach to ensure flexibility and scalability.

6.3.2 System Features

The system includes the following key modules:

- **Menu Management:**

Enables administrators to manage food items and their availability.

- **Sales Analytics:**

Provides insights into sales performance and customer preferences.

- **Demand Forecasting:**

Uses historical data to estimate future demand.

- **Online Tokenisation:**

Allows customers to place orders digitally, reducing physical queues.

6.3.3 System Architecture

The system follows a client-server architecture where the frontend handles user interaction, the backend processes logic, and the database stores all relevant data.

Chapter 7

QUEUING ANALYSIS

This chapter presents a detailed analysis of the existing canteen system using real-world data and queuing theory. It includes survey insights, calculation of system parameters, identification of the system as an M/G/1 model, and estimation of waiting time using the Kingman approximation, highlighting inefficiencies and the need for system improvement.

7.1 Introduction

This chapter presents the detailed queuing analysis of the canteen system at the College of Engineering Trivandrum. The analysis is based on data collected through field observation and processed using statistical tools. The objective is to model the existing system, evaluate its performance, and compare it with alternative configurations.

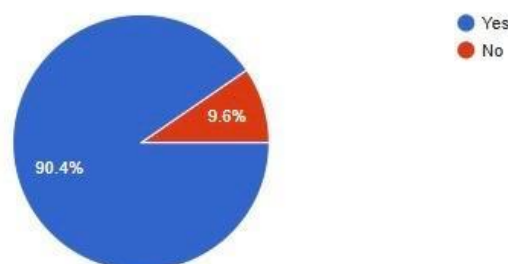
The study integrates survey insights, empirical data, and queuing theory to understand system behaviour and identify inefficiencies.

7.2 Survey Insights

A preliminary survey conducted among students revealed that a majority of users experience significant waiting times during peak hours. The results indicated dissatisfaction with the manual tokening system and overcrowding at the billing counter. These findings justified the need for a detailed queuing analysis and validated that waiting time is a critical performance metric for the system. Some of the responses are as follows

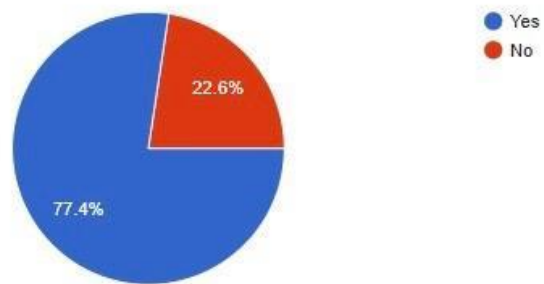
18. If you could pre-order (digital tokening) and pay for your food using a mobile app from your class, would you use it?

292 responses



6. If the line for ordering looks very long, are you more likely to leave and eat elsewhere?

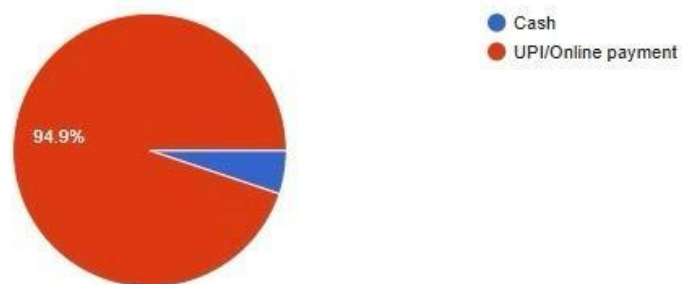
292 responses



:

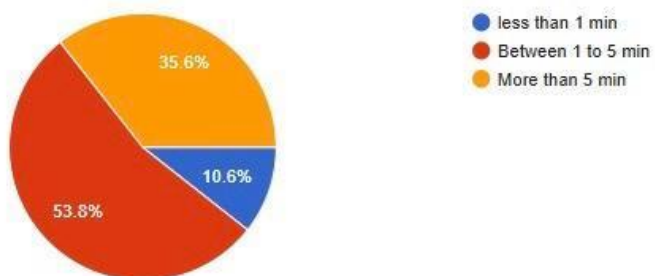
10. Which payment option do you prefer the most?

292 responses



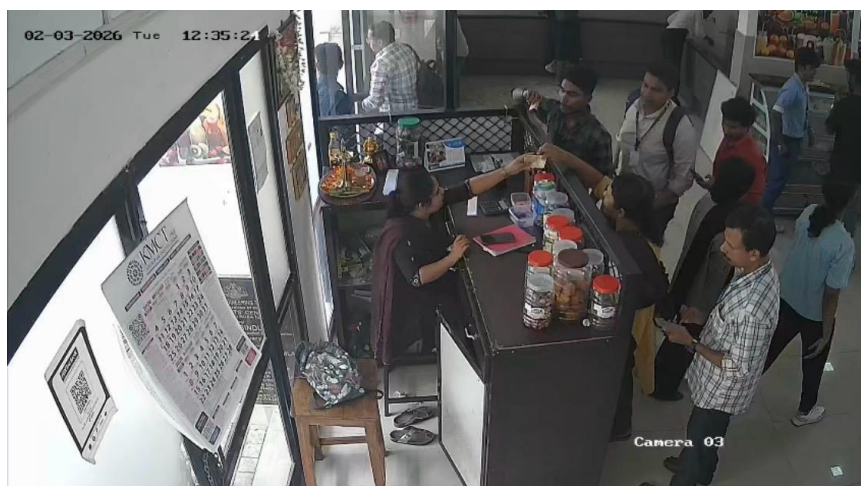
4. On average, how long do you feel you wait in line before you pay for your food?

292 responses



7.3 Data Collection and Processing

Primary data was collected through direct observation and through cctv footage of the billing counter of the canteen during peak hours.



The following parameters were recorded:

- Arrival time of each customer
- Inter-arrival time (time between consecutive arrivals)
- Service time at the billing counter

The collected data was entered into Microsoft Excel, where statistical analysis was performed.

7.4 Calculation of System Parameters

Using Excel, the following key metrics were calculated:

1. Mean Inter-arrival Time

$$\begin{aligned}\text{Mean Inter-arrival Time} &= \frac{\sum \text{Inter-arrival Times}}{n} \\ &= \mathbf{17.6395 \text{ seconds}}\end{aligned}$$

This represents the average time between customer arrivals.

2. Arrival Rate (λ)

$$\begin{aligned}\lambda &= \frac{1}{\text{Mean Inter-arrival Time}} \\ \lambda &= 1/17.63951 = \mathbf{0.0567 \text{ customers/sec}} \\ \lambda &= 0.0567 \times 60 = \mathbf{3.40 \text{ customers/min}}\end{aligned}$$

It represents the number of customers arriving per minute.

3. Mean Service Time

$$\begin{aligned}\text{Mean Service Time} &= \frac{\sum \text{Service Times}}{n} \\ &= \mathbf{12.96825 \text{ seconds}}\end{aligned}$$

This indicates the average time taken to serve a customer.

4. Service Rate (μ)

$$\begin{aligned}\mu &= \frac{1}{\text{Mean Service Time}} \\ &= 1/12.968251 = \mathbf{4.62 \text{ customers/min}}\end{aligned}$$

It represents how many customers can be served per minute.

5. Standard Deviation

Standard deviation was calculated for both inter-arrival time and service time to measure variability.

- Standard deviation of inter-arrival time = **17.928326 seconds**
- Standard deviation of service time = **9.170373 seconds**

6. Coefficient of Variation

$$C_v = \frac{\text{Standard Deviation}}{\text{Mean}}$$

- C_a : variability in arrival
- C_s : variability in service
- $C_a = \frac{17.928326}{17.6395} = \mathbf{1.016}$
- $C_s = \frac{9.170373}{12.96825} = \mathbf{0.707}$

7. Utilisation Factor (ρ)

$$\rho = \frac{\lambda}{\mu} = 0.736$$

This indicates how busy the system is.

- If $\rho < 1 \rightarrow$ system is stable
- If $\rho \geq 1 \rightarrow$ system is unstable

7.5 Identification of Queue Model

The canteen system is modelled as an **M/G/1 queue** based on its characteristics.

Queue Model Representation

$$A/S/m$$

Where:

- A = Arrival distribution
- S = Service distribution
- m = Number of servers

Meaning of M (Markovian / Exponential)

The arrival process is assumed to follow a **Poisson distribution**, meaning inter-arrival times are exponentially distributed.

This implies:

- Customers arrive randomly
- The process is memoryless

This assumption is reasonable for a canteen where arrivals are not scheduled but occur randomly during peak hours.

Meaning of G (General Distribution)

The service time follows a **general distribution**, meaning it does not strictly follow any specific distribution such as exponential.

This is because:

- Service time varies based on order complexity
- Payment delays and human interaction introduce variability

Hence, a general distribution is used for flexibility.

Meaning of 1 (Single Server)

The system has a single billing counter, making it a **single server system**.

Conclusion

Thus, the system is best represented as:

$$\mathbf{M/G/1}$$

7.6 Waiting Time Estimation (Current System)

The average waiting time in queue is estimated using the **Kingman Approximation**:

The average waiting time in queue is estimated using the Kingman Approximation (VUT formula). This method is particularly suitable for real-world systems where both arrival and service processes exhibit variability.

Unlike simpler queuing models that assume exponential or constant distributions, the Kingman approximation incorporates the coefficient of variation of inter-arrival and service times, making it more accurate for practical scenarios such as canteen operations.

In this study, the collected data shows significant variability in both arrival and service processes ($C_a \neq 0$, $C_s \neq 0$). Therefore, the Kingman approximation is chosen as it effectively captures the impact of this variability on waiting time.

$$W_q = \frac{\rho}{1 - \rho} \cdot \frac{C_a^2 + C_s^2}{2} \cdot \frac{1}{\mu}$$

Substitution of Values

$$\begin{aligned}\rho &= 0.736 \\ \frac{\rho}{1 - \rho} &= \frac{0.736}{0.264} = 2.79 \\ C_a^2 &= (1.016)^2 = 1.033 \\ C_s^2 &= (0.707)^2 = 0.500 \\ \frac{C_a^2 + C_s^2}{2} &= 0.7665\end{aligned}$$

$$\frac{1}{\mu} = 12.96825 \text{ seconds}$$

Final Result

$$W_q = 2.79 \times 0.7665 \times 12.96825$$

$$W_q \approx 27.6 \text{ seconds}$$

$$W_q \approx 0.46 \text{ minutes}$$

7.7 Interpretation of Results

The results indicate that the current system is stable but experiences noticeable waiting time due to variability in arrivals and service processes.

The relatively high coefficient of variation of arrival ($C_a > 1$) indicates significant randomness in customer arrivals, which contributes to queue formation. Additionally, the system operates at a moderate utilisation level, which increases waiting time during peak conditions.

As the utilisation factor approaches 1 during rush periods, the waiting time is expected to increase sharply, leading to congestion.

7.8 Need for System Improvement

The analysis highlights the limitations of the current single-server system in handling peak demand efficiently. Therefore, alternative service configurations such as:

- Two-parallel-billing system
- Two-dedicated-billing system

are analysed in subsequent sections to evaluate improvements in performance.

Chapter 8

MODELLING AND SIMULATION

This chapter evaluates different service configurations through modelling and simulation, including the current single-server system, two-parallel-billing system, and two-dedicated-billing system. Using modified queuing equations and Python-based simulation, it compares performance and demonstrates that the two-parallel system significantly reduces waiting time.

8.1 Introduction

This chapter presents the modelling and simulation of the canteen queuing system under different configurations. Based on the data collected and analysed in the previous chapter, three models are considered:

1. Current system (single-server)
2. Two-parallel-billing model
3. Two-dedicated-billing model

The objective is to evaluate and compare the performance of these models in terms of average waiting time, particularly under varying arrival conditions.

8.2 Modelling Approach

The canteen system is modelled using queuing theory concepts. Since real-world systems exhibit variability in both arrival and service processes, the **Kingman Approximation (VUT formula)** is used as the basis for modelling.

The general form of the equation is:

$$W_q \approx \frac{\rho}{1 - \rho} \cdot \frac{C_a^2 + C_s^2}{2} \cdot \frac{1}{\mu}$$

However, for multi-server systems, a modified form of the equation is required.

8.3 Modification of Kingman Equation for Multi-Server Systems

For systems with multiple servers ($m > 1$), the standard Kingman formula for single-server queues is extended using an approximation.

The modified form is:

$$W_q \approx \frac{C_a^2 + C_s^2}{2} \cdot \frac{\rho^{\sqrt{2(m+1)}-1}}{m(1-\rho)} \cdot \frac{1}{\mu}$$

Why this modification is used

- The original Kingman equation is strictly applicable to **single-server (G/G/1) systems**
- The canteen improvement involves **multiple servers (m = 2)**
- Exact solutions for G/G/m systems are complex
- This approximation provides:
 - Practical accuracy
 - Computational simplicity
 - Ability to compare different configurations

Thus, this modified equation is used to estimate waiting time for multi-server models.

8.4 Model 1: Current System (Single Server)

The current system consists of a single billing counter and is modelled as an **M/G/1 queue**.

- Number of servers: $m = 1$
- Customers are served sequentially

The standard Kingman equation is used for this model.



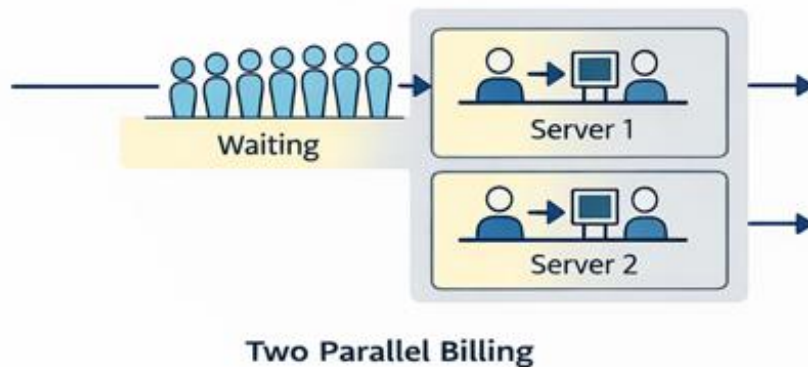
8.5 Model 2: Two-Parallel-Billing System

In this model, two billing counters serve a **common queue**.

- Number of servers: $m = 2$
- Customers are assigned to the first available server

Characteristics

- Reduced waiting time due to load sharing
- Pooling effect reduces variability impact

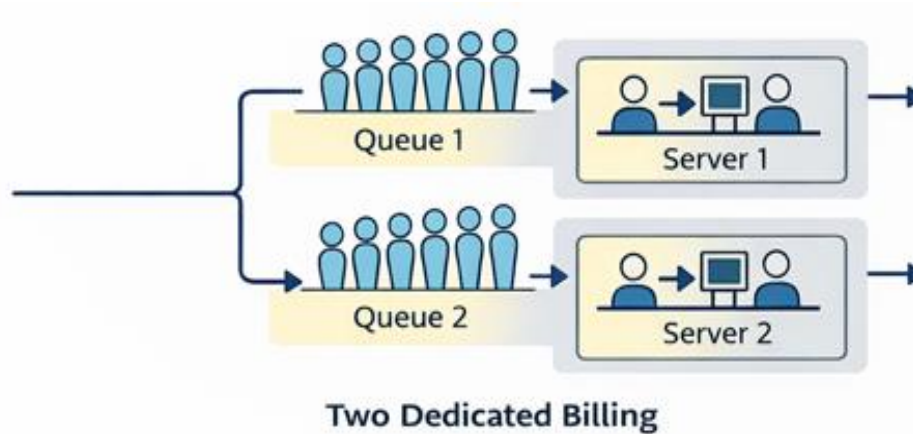


8.6 Model 3: Two-Dedicated-Billing System

In this model, the arrival stream is divided into two independent queues.

- Each server handles half of the arrivals

- Each queue behaves as an independent M/G/1 system



Characteristics

- No pooling advantage
- Higher sensitivity to variability

8.7 Simulation Setup

The simulation study was carried out using **Python**, with the help of the following libraries:

- **NumPy** – for numerical computations
- **Pandas** – for data handling
- **Matplotlib** – for generating graphs

The simulation was performed by varying the **coefficient of variation of inter-arrival time (C_a)** over a range from 0 to 2.

The **coefficient of variation of service time (C_s)** was assumed to be 0, representing a deterministic service process. This assumption simplifies the analysis and allows the study to focus on the effect of arrival variability on system performance.

For each value of C_a , the corresponding waiting time was computed for all three models using the respective equations.

QUEUEING THEORY - SIMULATION

```
import pandas as pd
import math
import numpy as np
import matplotlib.pyplot as plt
```

Parameters

```
# Arrivals
ra = 3.401 # customer/min
ta = 1/3.401 # min/customer
# Processing
rp = 4.626683 # customer/min
tp = 1/4.626683 # min/customer
```



```
# Create list for result
x,y_sys, y_1, y_2 = [], [], [], []
for p in np.linspace(0,2,2*10+1):
    CVA = p
    # CVP value
    x.append(CVA)
    # tq calculation
    y_sys.append(tq_sys(ra, ta, rp, tp, CVA, CVP))
    y_1.append(tq_s1(ra, ta, rp, tp, CVA, CVP))
    y_2.append(tq_s2(ra, ta, rp, tp, CVA, CVP))

#multiplication with 60
y_sys_min = [m * 60 for m in y_sys]
y_1_min = [m * 60 for m in y_1]
y_2_min = [m * 60 for m in y_2]

# Results
df_result = pd.DataFrame({'CVA': x, 'crnt_sys': y_sys_min, 'SOL1':y_1_min, 'SOL2':y_2_min})
df_result['Δ(2,1)'] = df_result['SOL2'] - df_result['SOL1']
df_result.tail()
```

Scenario: Deterministic Service Distribution

$CV_p = 0$

```
▶ # Deterministic Distribution for service
CVP = 0
x, y_sys, y_1, y_2 = [], [], [], []
CVA = 0.1
# CVP value
x.append(CVA)
# tq calculation
y_sys.append(tq_sys(ra, ta, rp, tp, CVA, CVP))
y_1.append(tq_s1(ra, ta, rp, tp, CVA, CVP))
y_2.append(tq_s2(ra, ta, rp, tp, CVA, CVP))
```

Function to simulation solutions

```
# Solution 1
def tq_s1(ra, ta, rp, tp, CVA, CVP):
    u = 1/2 * ra/rp
    tq = ((CVA**2+CVP**2)/2)*((math.pow(u, math.sqrt(6)-1)))/(2*(1-u))*tp
    return tq

# Solution 2
def tq_s2(ra, ta, rp, tp, CVA, CVP):
    u = 1/2 * ra/rp
    tq = ((CVA**2+CVP**2)/2)*((u)/(1-u))*tp
    return tq

def tq_sys(ra, ta, rp, tp, CVA, CVP):
    # Utilization (u) = arrival rate / service rate
    u = ra / rp

    # Kingman Formula: V * U * tp
    # V = (CVA^2 + CVP^2) / 2
    # U = u / (1 - u)
    tq = ((CVA**2 + CVP**2) / 2) * (u / (1 - u)) * tp

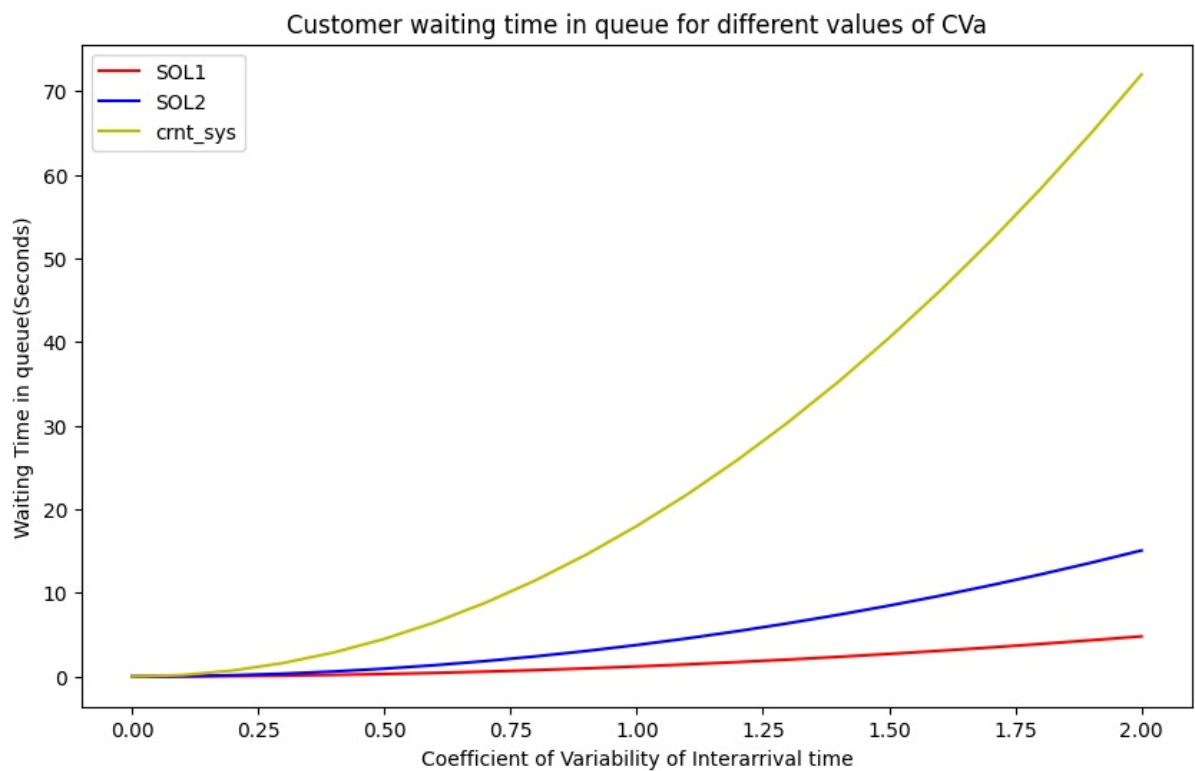
    return tq
```

8.8 Results and Graphical Analysis

The results of the simulation are represented using a graph of:

- **X-axis:** Coefficient of variation of inter-arrival time (C_a)
- **Y-axis:** Average waiting time (W_q)

	CVa	crnt_sys	SOL1	SOL2	$\Delta(2,1)$
0	0.0	0.000000	0.000000	0.000000	0.000000
1	0.1	0.179920	0.012015	0.037681	0.025667
2	0.2	0.719681	0.048058	0.150725	0.102667
3	0.3	1.619282	0.108131	0.339132	0.231002
4	0.4	2.878724	0.192232	0.602902	0.410670
5	0.5	4.498006	0.300363	0.942034	0.641671
6	0.6	6.477128	0.432523	1.356529	0.924006
7	0.7	8.816091	0.588712	1.846387	1.257675
8	0.8	11.514894	0.768930	2.411608	1.642678
9	0.9	14.573538	0.973176	3.052191	2.079014
10	1.0	17.992022	1.201452	3.768137	2.566684
11	1.1	21.770347	1.453757	4.559446	3.105688
12	1.2	25.908512	1.730091	5.426117	3.696026
13	1.3	30.406518	2.030455	6.368151	4.337697
14	1.4	35.264364	2.354847	7.385548	5.030701
15	1.5	40.482050	2.703268	8.478308	5.775040
16	1.6	46.059577	3.075718	9.646430	6.570712
17	1.7	51.996944	3.472197	10.889915	7.417718
18	1.8	58.294152	3.892706	12.208763	8.316057
19	1.9	64.951200	4.337243	13.602974	9.265731
20	2.0	71.968089	4.805810	15.072547	10.266738



Observations

- The **current system** shows a steep increase in waiting time as variability increases
- The **two-dedicated system** also exhibits significant growth in waiting time
- The **two-parallel system** maintains relatively low waiting time across all values of C_a

8.9 Interpretation of Results

The results demonstrate that:

- Waiting time is highly sensitive to arrival variability, particularly in systems with limited service capacity.
- The **current system** shows a rapid increase in waiting time as the variability in arrivals increases, indicating poor performance under peak conditions.
- The **two-dedicated system**, although better than the current system, still exhibits significant sensitivity to variability due to the absence of resource sharing between servers.

The **two-parallel-billing system** consistently outperforms both other configurations due to the pooling of servers, which reduces the impact of variability and improves service efficiency.

Performance Comparison

- The **two-parallel-billing system reduces waiting time by approximately 93% compared to the current system**, demonstrating a substantial improvement in performance.
- It also achieves around **68% reduction in waiting time compared to the two-dedicated-billing system**, highlighting the advantage of shared service channels over independent queues.

8.10 Conclusion

The analysis reveals that both the current system and the two-dedicated-billing model are highly sensitive to variability in inter-arrival times. Any delay in payment or service will paralyse an entire queue in the two-dedicated model (50% of the input flow) and 100% of the input flow in the current system. In contrast, the two-parallel-billing model benefits from pooling: variability in one server does not block the other.

This demonstrates that the two-dedicated-billing model and the current system are suboptimal when inter-arrival time is highly variable — a common real-world scenario during peak hours.

Chapter 9

ONLINE ORDER MANAGEMENT SYSTEM

9.1 Introduction

This chapter presents the design and implementation of the Online Order Management System developed as part of this project. The system is proposed as a digital solution to overcome the inefficiencies observed in the existing manual canteen operation.

In the current system, customers are required to physically stand in queues to place orders and receive tokens. This leads to overcrowding, delays, and inefficient utilisation of service time. The developed system introduces an online interface through which customers can place orders in advance, thereby reducing congestion and improving overall service efficiency.

Additionally, the system enables the collection and storage of transaction data, which can be further utilised for analytics and demand forecasting. Thus, the system not only improves operational efficiency but also supports data-driven decision-making.

9.2 Objectives of the System

The Online Order Management System is designed with the following objectives:

- **Reduction of waiting time:** By enabling customers to place orders digitally, the system reduces the need for physical queuing and minimizes waiting time at the billing counter
- **Automation of Tokening Process:** The system replaces manual token issuance with automated digital token generation, eliminating delays and human errors.
- **Digital Record Maintenance:** All transactions are recorded in a structured database, allowing easy retrieval and analysis of data.
- **Real-Time Monitoring:** Administrators can track incoming orders and manage them efficiently in real time.

- **Support for Analytics and Forecasting:** The stored data is used to analyse demand patterns and predict future requirements, improving inventory and production planning.

9.3 Technology Stack

The system is developed using **Python** as the primary programming language, along with the **Flask** framework for backend development.

Frontend Technologies

HTML: Used to structure the web pages and define the layout of the interface.

CSS: Used for styling and improving the visual appearance of the application.

JavaScript: Used to add interactivity, handle user inputs, and dynamically update content.

Backend Technologies

FlaskFramework:

Handles routing, request processing, and server-side logic. It connects the frontend interface with the database and processes user actions such as placing orders.

Database

The database stores:

- Menu details
- Order information
- Transaction records

MySQL is used here

Supporting Libraries

Pandas

Used for handling and processing sales data for analytics and forecasting.

NumPy

Used for performing numerical computations required in data analysis.

Matplotlib

Used for generating graphical representations of sales data and trends.

9.4 System Architecture

The system follows a **three-tier client-server architecture**, which separates the application into distinct layers:

1. Presentation Layer (Frontend)

This layer is responsible for user interaction. It provides interfaces for customers and administrators to interact with the system.

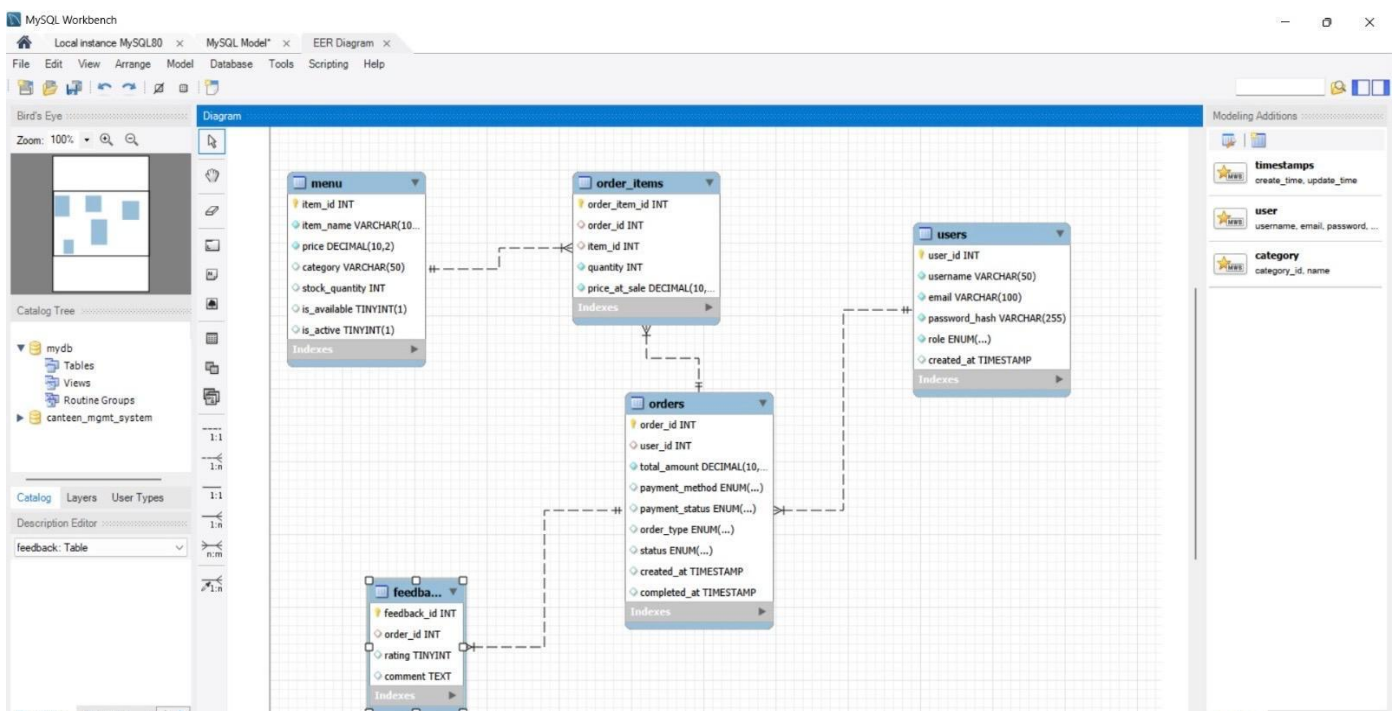
2. Application Layer (Backend)

This layer processes requests received from the frontend. It handles business logic such as order processing, token generation, and data validation.

3. Data Layer (Database)

This layer manages data storage and retrieval. It stores all relevant information, including orders, menu items, and transaction records.

This layered architecture ensures modularity, scalability, and ease of maintenance.



9.5 System Modules

The system is divided into multiple modules, each responsible for a specific functionality.

9.5.1 Menu Management Module

This module is used by the administrator to manage the menu displayed to customers.

Functions:

- Adding new food items
- Updating prices and availability
- Removing items that are out of stock

Significance:

This module ensures that customers always view an updated and accurate menu, which improves the ordering experience and prevents confusion.

Admin Panel

+ Add New Menu Item

Item Name

Price (₹)

Lunch

Add Item

Item Details	Price	Stock Status	Manage
Masala Dosa Breakfast	₹50.00	In Stock Out	Edit
Idily Breakfast	₹35.00	In Stock Out	Edit
Chappati w/ Kadala Curry Breakfast	₹40.00	In Stock Out	Edit
Egg Curry Breakfast	₹25.00	In Stock Out	Edit
Biriyani Lunch	₹70.00	In Stock Out	Edit
Meals Lunch	₹30.00	In Stock Out	Edit
Chicken Curry/Fry Lunch	₹50.00	In Stock Out	Edit
Fish Curry/Fry Lunch	₹60.00	In Stock Out	Edit

MENU

TOKEN

SATES

MRP

- This enables administrators to efficiently manage all food items available in the canteen. It provides a simple and dynamic interface to add, update, and control menu items in real time.
- It also includes a stock status feature, which helps in marking items as **In Stock** or **Out of Stock**, ensuring that unavailable items are not displayed for ordering.
- Any changes made are instantly reflected across the entire application, including the order management and sales analytics modules.

9.5.2 Order Management Module

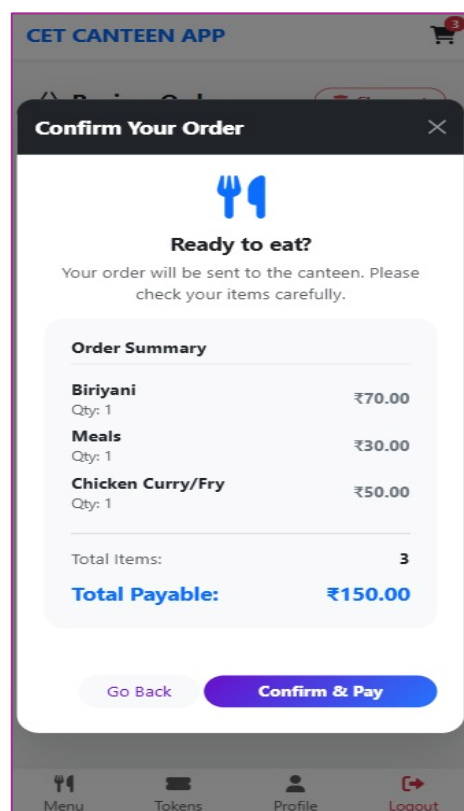
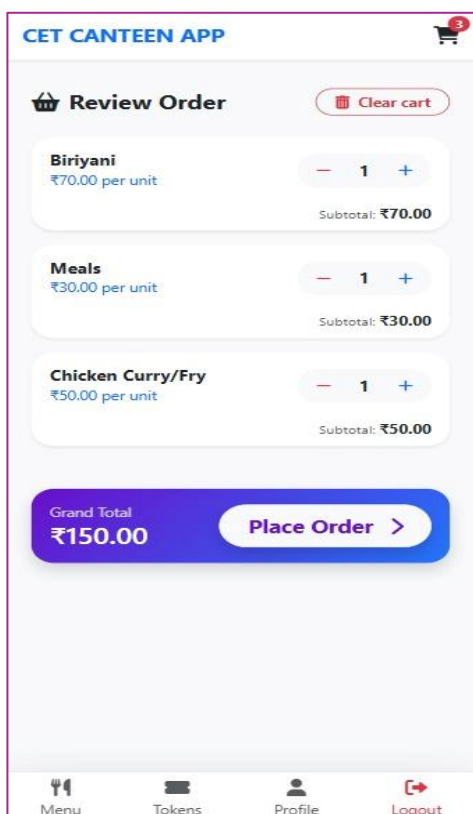
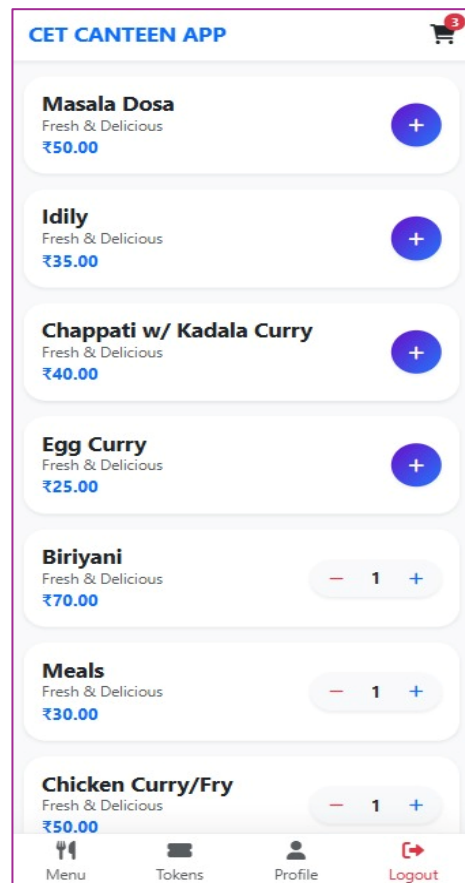
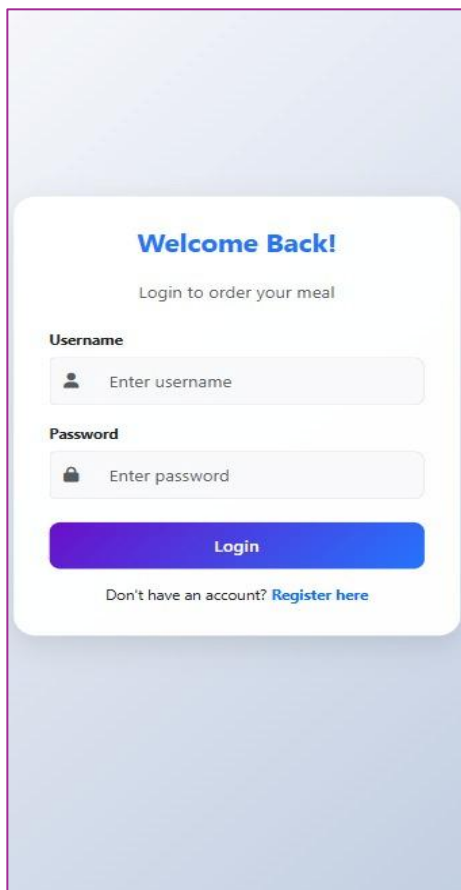
This is the core component of the system responsible for handling customer orders.

Functions:

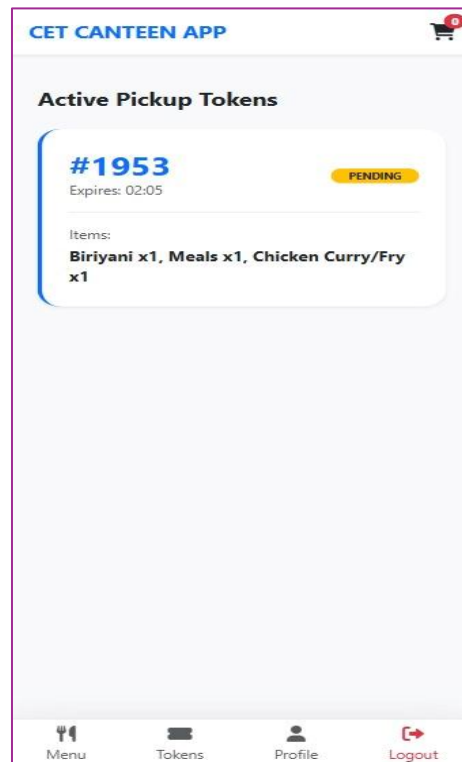
- Accepting orders from users
- Processing order details
- Storing order information in the database

Significance:

It replaces the manual order-taking process, ensuring faster and more efficient handling of customer requests.



9.5.3 Token Generation System



This module generates a unique token number for each order placed.

Functions:

- Assigning sequential token numbers
- Displaying token details to customers
- Tracking order status

Significance:

It eliminates manual token distribution and reduces crowding near the billing counter.

9.5.4 Sales Analytics Module

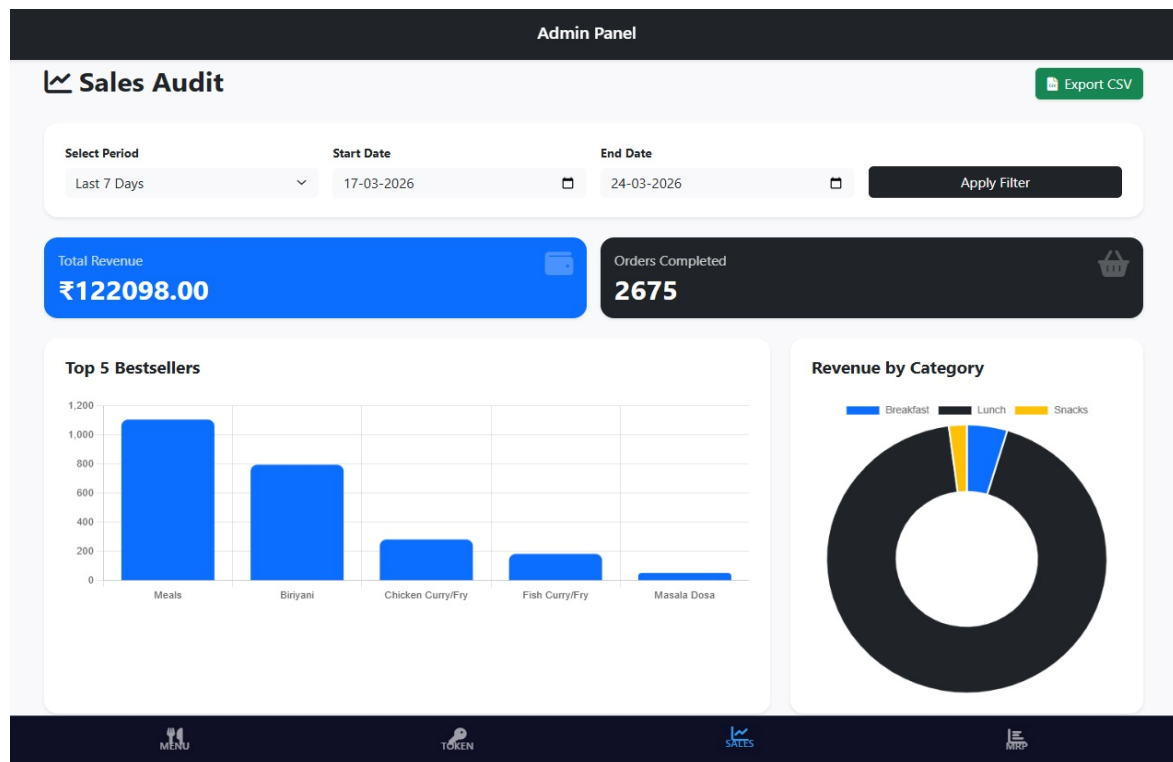
This module processes transaction data to provide useful insights.

Functions:

- Generating sales reports
- Identifying frequently ordered items
- Tracking revenue trends

Significance:

It helps management understand demand patterns and make informed decisions regarding inventory and pricing.



It also highlights the top-selling items and provides a category-wise revenue breakdown (Breakfast, Lunch, Snacks) for better understanding of demand distribution.

9.5.5 Demand Forecasting Module

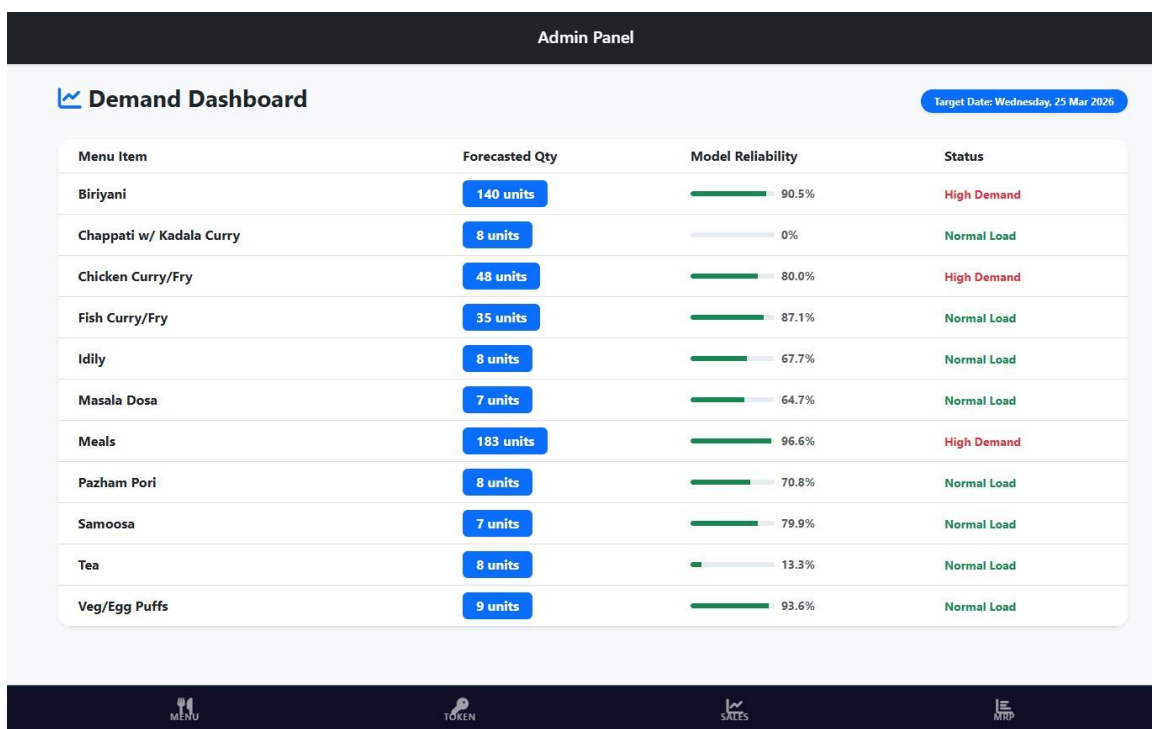
This module uses historical sales data to estimate future demand.

Functions:

- Analysing past order trends
- Predicting future demand for items
- Supporting production planning

Significance:

It helps reduce food wastage and prevents stockouts by improving demand estimation.



Demand forecasting is currently implemented using META's prophet model

9.6 System Workflow

The system workflow describes the complete sequence of operations involved in processing a customer order, from initial interaction to final data storage and analysis. The workflow ensures smooth coordination between the customer interface, backend processing, and administrative operations.

Step 1: User Access and Interface Loading

The user accesses the Online Order Management System through a web browser. The frontend interface is loaded dynamically, displaying the available menu items retrieved from the database.

- The system ensures that only currently available items are displayed
- The interface is designed to be user-friendly and responsive

Step 2: Menu Display and Item Selection

The menu is presented to the user in a structured format, including:

- Item names
- Prices
- Availability status

The user selects the desired food items and specifies the required quantity for each item.

- The system updates the total cost dynamically
- Input validation ensures correct quantity selection

Step 3: Order Placement

Once the selection is complete, the user proceeds to place the order.

- The selected items and quantities are sent to the backend server
- The system verifies the availability of items before confirming the order
- Any invalid or unavailable items are flagged for correction

Step 4: Order Processing in Backend

The backend system processes the order using the application logic.

- Order details are structured and formatted
- A new order entry is created
- The system prepares the order for token generation

This step ensures that the order data is consistent and ready for storage.

Step 5: Token Generation

A unique token number is generated for the order.

- Tokens are assigned sequentially
- Each token acts as an identifier for order tracking
- The token is displayed to the user upon successful order placement

This replaces the traditional manual tokening system.

Step 6: Database Storage

The order details are stored in the database.

Stored information includes:

- Token number
- Items ordered
- Quantity
- Total cost
- Timestamp of order

This creates a permanent digital record of transactions.

Step 7: Order Queue Management

All incoming orders are maintained in a queue within the system.

- Orders are processed in the order of arrival (FIFO principle)
- The queue is accessible to the administrator
- The system ensures orderly processing of orders

Step 8: Admin Interface and Order Handling

The administrator accesses a separate interface to manage orders.

- Displays all pending and completed orders
- Allows marking orders as “in progress” or “completed”
- Helps coordinate preparation and delivery

This improves operational efficiency inside the canteen.

Step 9: Order Completion and Delivery

Once the order is prepared:

- The corresponding token number is used to identify the customer
- The order is handed over to the user
- The system updates the order status as completed

Step 10: Data Recording for Analytics

All transaction data is automatically recorded for further analysis.

- Sales data is stored for analytics
- Item-wise demand is tracked
- Data is used for generating reports and forecasting

This step enables the transition from manual operation to a data-driven system.

9.7 Advantages of the System

- Reduces waiting time and physical congestion
- Improves service speed and efficiency
- Minimizes human errors in billing and tokening
- Provides real-time data for decision-making
- Enhances overall user experience

9.8 Limitations

- The system is currently a prototype and not deployed in a real-time production environment
- Forecasting module uses basic techniques and can be further improved

9.9 Conclusion

The Online Order Management System demonstrates how digital technologies can be effectively applied to improve the efficiency of institutional service systems. By automating order processing and integrating analytics, the system addresses both operational and managerial challenges.

The implementation of such a system in a real-world environment has the potential to significantly enhance the performance of the canteen and improve customer satisfaction.

Chapter 10:

CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

This project successfully addressed the dual challenges of long waiting times and poor data management in the canteen at the College of Engineering Trivandrum. Through a rigorous application of queuing theory and the Kingman Approximation, three service models were analysed and compared. The two-parallel-billing model emerged as the clearly superior configuration, offering a near 93% reduction in predicted waiting time with minimal implementation overhead.

Simultaneously, an Online Order Management System was developed using the Python Flask framework, featuring three integrated modules: Menu Management, Sales Analytics, and Demand Forecasting. The system provides the canteen management with real-time visibility into sales data and actionable demand predictions, enabling better resource planning and waste reduction.

Together, these contributions demonstrate that industrial engineering methodologies and modern software tools can be effectively combined to transform the efficiency of institutional canteen operations.

10.2 Limitations

- The queuing analysis is based on data collected over a limited observation period and may not capture seasonal variations in demand.
- The demand forecasting module requires a sufficient history of sales data to produce accurate predictions; accuracy may be low in the initial deployment phase.
- The software system is a prototype and has not been subjected to full-scale production deployment or security hardening.

10.3 Future Scope

- Integration of the online tokenisation system with a mobile application for greater accessibility.
- Deployment of the Online Order Management System in the actual canteen with real-time monitoring.
- Application of more advanced forecasting techniques (e.g., LSTM neural networks) for improved demand prediction accuracy.
- Extension of the queuing analysis to include multi-stage queues (ordering, payment, food collection) for a more comprehensive model.
- Exploration of dynamic pricing or pre-order discounts to spread peak demand across time.

Chapter 10:

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